

Energy consumption of running an AI?

Posted by u/Naeio_Galaxy • 0 points • 5 comments

I have a technical question. Most of us all already know that training an AI uses a lot of energy. My question isn't however on training an AI, it's about running it. Once you get a trained AI, does it take quite a bit of energy to run it, or is more modest quantities? Can we even compare it in terms of memory usage and performances to apps on regular desktops or phones? (Btw talking in execution speed instead of energy consumption is fine by me)

I guess the energy consumption is dependent on the memory size of the AI, and honestly I don't know much on the subject, so if you can say how execution speed scales with memory usage it would be great.

Comments

u/LcuBeatsWorking • 1 points • 2023-05-11 12:24 UTC

>Once you get a trained AI, does it take quite a bit of energy to run it, or is more modest quantities?

What means "running an AI"? Something like ChatGPT as a service? Or something like Siri?

>I guess the energy consumption is dependent on the memory size of the AI

Not sure what you mean by that. Energy usage is mainly determined by CPU and GPU power consumption, not memory. Holding data in memory does not need much energy (which is why your laptop can suspend for a very very long time).

u/Naeio_Galaxy • 1 points • 2023-05-13 13:41 UTC

>What means "running an AI"? Something like ChatGPT as a service? Or something like Siri?

What's the difference between them that you wanna emphasis?

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I thought of it more as a "cost per request". The bigger the AI, the more things there are to compute I guess (and thus the longer for 1 request, and running CPU/GPU for longer means more energy). But it's just a guess

u/Cerulean_IsFancyBlue • 1 points • 2023-05-10 22:03 UTC

It's a very interesting question, and you might be able to do a fermi estimate.

I think when it comes to an accurate answer, you're going to need to refine the question, and have access to some internal statistics that you're not going to find asking Reddit.

u/noideeewhat • 1 points • 2023-05-10 20:19 UTC

Under 20w at idle.

I don't really think people host some gpt like AIs at home and your question it's not that simple to answer.

u/AutoModerator • 1 points • 2023-05-10 17:23 UTC

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- * AI is going to take our jobs - its been asked a lot!
- * Discussion regarding positives and negatives about AI are allowed and encouraged. Just be respectful.
- * Please provide links to back up your arguments.
- * No stupid questions, unless its about AI being the beast who brings the end-times. It's not.

Thanks - please let mods know if you have any questions / comments / etc

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